

# Final Year Thesis Report

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A thesis submitted to the Department of Computer Science and Engineering  
in partial fulfillment of the requirements for the degree of  
B.Sc. in Computer Science and Engineering

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# Declaration

It is hereby declared that

1. The thesis submitted is my/our own original work while completing degree at Brac University.
2. The thesis does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
4. We have acknowledged all main sources of help.

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# Approval

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# Abstract

Traditional authentication mechanisms primarily rely on static passwords, which may compromise security when credentials are stolen or exposed. This research examines continuous user verification using non-intrusive, free-text keystroke dynamics, which is the verification of users based on their natural typing behavior. The proposed method is based on the Zero Trust Architecture (ZTA) philosophy of “never trust, always verify,” which involves continuously evaluating user identity throughout active sessions without the need for repeated authentication. Key hold time and inter-key flight latency are evaluated to recognize user-specific behavioral patterns and to detect unauthorized access. Unique typing patterns are identified using machine learning algorithms from benchmark keystroke data sets. This work bridges the gap between security and usability, and the potential of creating secure computing experiences for a user is discussed. Additionally, sensitive personal data is protected from unauthorized transmission. In the end, this research offers an effective and sustainable framework that paves the way for real-world implementation in individual devices and secure metaverse environments.

**Keywords:** Keyboard dynamics, Flight Time, Free-Text Typing, Hold Time, ZTA, FAR, FRR ,Catastrophic Forgetting,ERR, GNN.

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# Chapter 1

## Introduction

### 1.1 Background

Continuous Authentication verifies a user throughout the session, and this is becoming increasingly critical as passwords, PINs and static credentials are easy to steal, spoof, brute force, and session steal [18]. Keystroke dynamics is one of the most researched behavioural biometric modalities to be used for this purpose as users have unique typing patterns which can be used with a standard keyboard or with touch input without requiring any special hardware or user effort [19], [20]. Continuous authentication has also been explored in the wider literature not only for keystroke dynamics but for mouse dynamics, touch, gait and multimodal combinations, which are all passive and user-friendly in real-world systems. [7], [20]. Recent model driven studies have reported high performance of keystroke authentication for deep ensembles, RL, Siamese networks, LightGBM, optimized BRNNs, and LSTM based systems with reported accuracy ranging from 80% to 99% depending on the dataset and task definition [4].

### 1.2 Rational of the Study or Motivation

The primary reason for this study is that static authentication is not enough for a zero-trust environment, as identities need to be continuously updated to minimize identity theft, insider misuse and hijacking after login [14], [15]. The included zero-trust papers extend the argument that users' authentication should not be separated from adaptive access control as many existing systems currently authenticate the device or user but do not tightly support continuous trust reassessment in enterprise workflows. [15], [18]. The second is that keystroke biometrics are deployable since they can be based on the existing human-computer interaction data and enhance security without increasing user burden. [17]. A third motivation is that recent research has shown that these systems can become increasingly effective in the longer term, such as methods for short sequences, uncontrolled environments, continual adaptation, multimodal fusion, and efficient on-device or privacy-preserving execution using machine learning [3], [7], [18].

## 1.3 Problem Statement

While all these results are promising, existing behavioral-biometric continuous authentication systems have still not overcome a number of limitations. Behavioral-based solutions may not yet be accurate in real-world situations, may have false acceptance and false rejection failures, and are still susceptible to attacks like replay, mimicry, template leakage and synthetic or forged behavior.[5], [19]. User behaviour changes with each session, scene, device and environment, degrading the performance of the model and reducing the possibility of generalisation; adaptation, cross-scenario transfer and robustness in uncontrolled environments are open problems, identified in several attached papers [3], [20]. There are also some areas where improvements are needed, such as the reliance on a single modality, a lack of standards for enrollment and lockout behavior, inadequate privacy protection, and poor integration with scalable zero-trust architectures. [5], [10], [18]

## 1.4 Objective

The overall purpose of this study is to create a machine-learning based continuous keystroke authentication system to check whether the user who is typing on the keyboard is the legitimate user or not during an ongoing session in a closed environment with no trust assumptions. The study must be to:

- Extract discriminative features of keystrokes like sequence, flight time, hold time and related behavioural features for user verification.
- Develop and test an effective classifier for GU-Impostor detection based on recent ML / DL techniques appropriate for continuous authentication.
- Design with robustness to short inputs, changing behaviour, and real-world variation, either through adaptation or continual learning; which is best for each use case?
- align the system to privacy preserving and zero-trust deployment requirements, to continuously re-evaluate identity for adaptive access control with minimal user friction.

## 1.5 Methodology in Brief

The structure verifies the identity of the user every time through executing five basic procedures: acquisition, preprocessing, extraction, categorization, and evaluation.

**1.Acquisition:** This initiates from gathering data from everyday devices with sensors based on the keystroke event data that is captured from normal typing without any specific authentication action. Key press and release times, key codes, and other relevant data may be included. Short sequences or temporal windows are used as inputs for continuous user verification.

**2.Preprocessing:** The recorded keystroke data is cleansed of invalid, incomplete, or unusual records. Key-press and key-release events are synchronized to obtain

meaningful temporal measurements. Events are processed and organized in appropriate sequences or temporal windows, and numerical features are normalized where appropriate. Training, validation and testing data are treated in a uniform way to minimize data leakage.

**3.Extraction:** There are core features, such as hold time (dwell time) and variants of flight-time as Press-to-Press, Release-to-Press, Press-to-Release or Release-to-Release. Random Forest takes these as tabular statistics per window (mean/variance of timings, digraph latencies). The 1D CNN learns local rhythm patterns automatically from raw timing sequences as 1D signals. The BiLSTM is forward and backward bidirectional to learn long-range, directionally sensitive temporal dependencies in typing rhythm by processing sequences of key-embedding and timing.

**4.Model Training:** All three models do closed-set continuous verification, which means they tell the difference between enrolled users. Random Forest: Classifies tabular feature vectors. 1D CNN: Learns discriminative patterns end-to-end from raw sequences, especially for short and uncontrolled sequences, where it learns context. Each model scores a classification or confidence score, similar to what you would see in a real-time trust score in a Zero Trust setup.

**5.Evaluation :**At the last stage the system’s performance is examined. It monitors whether or not it’s performing by considering aspects including accuracy, the False Acceptance Rate (FAR), the False Rejection Rate (FRR) and the Equal Error Rate (EER). The system can add fresh users without starting over from scratch.

## 1.6 Scopes and Challenges

The scope of this research is closed-set continuous user verification in non-intrusive keystroke dynamics in the Zero Trust security context. It explores the timing-based behavioral features of normal typing . The study investigates whether these models can learn unique typing styles and if they can accurately predict whether the observed style is a legitimate user or not at all times.

**Challenges Intra User Variability :**This refers to the fact that typing behavior will change from one session to another and one situation to another, which can lead to more false rejection.

**Cross-Scenario and Hardware Variability:** The differences in the keyboard, device, environment or usage context can impact on model generalization.

**Limited Keystroke Sequences:** There may be insufficient behavioral information for reliable verification in shorter sequences of keystrokes.

**Representation Differences:** The four models use different representations for the input such as tabular features, sample pairs and graph structures, which makes it difficult to compare the models.

**Privacy:** Continuous authentication should not rely on the actual typed content, but on the timing characteristics of the behavior, to reduce the exposure of sensitive information.

**Temporal Variability:** Typing patterns can vary with time and models must be able to verifiably detect these changes in typing patterns across different typing sessions.

# Chapter 2

## Literature Review

### 2.1 Preliminaries

Biometric authentication systems can be broadly classified into physiological biometrics (example: fingerprint, iris geometry, and facial recognition) and behavioral biometrics (example: typing rhythm, cursor movements, touch screen gestures, and gait pattern). [5] Physiological biometrics are used to assess a person's static physical characteristics, whereas behavioral biometrics are used to assess a person's temporal, neurophysiological, and neuromuscular habits developed in human-computer interactions[5], [20]. Keystroke dynamics quantifies the motion and pressure of typing on a physical keyboard or touchscreen device, as well as the timing and rhythm of typing, using a unique motion pattern for each person. The unique motion pattern, timing and rhythm of each person typing on a physical keyboard or touchscreen device constitutes keystroke dynamics. Keystroke biometrics is unlike traditional knowledge based credentials (passwords, PINs) in that it is limited to the manner in which a document is typed and not what text is typed. Whereas traditional knowledge based credentials (passwords or PINs) are based on the text typed, keystroke biometrics is based solely on the way a document is typed. This is a hardware-free, cost-effective layer of continuous security that utilizes existing input hardware[1], [4].

**Static vs. Continuous Authentication Paradigms** Keystroke dynamics authentication modalities have two paradigms: Static Authentication (Fixed-Text): Authenticates the user's typing behavior at the start of a session against a known string of characters, for example, a login password or PIN[1], [12]. Static authentication is effective for initial access but it fails to recognize post login session hijacking and any unauthorized insider access[15], [18]. Continuous Authentication (Free-Text): Passively and transparently observes unconstrained typing activity during a digitally active session[5]. Continuous systems calculate user verification scores continuously and always enforce the Zero-trust principle, which doesn't create authentication fatigue[15], [18].

**Temporal and Key Signal Formulations.** Raw keystroke streams are sequences of key-press (P) and key-release (R) times, corresponding to successive keys  $K_i$  and  $K_{i+1}$ [14] that have been pressed and then unpressed. Hold Time / Dwell Time (HT): The amount of time a key  $K_i$  is pressed before being released, given by  $HT_i = R[i] - P[i]$ [14]. The time from the last key action to the next key action [14]: Press-to-Press (P2P / DD):  $P[i+1] - P[i]$ [15], [19] Release-to-Press (R2P / UD / Flight Time):  $P[i+1] - R[i]$ [14] Press-to-Release (P2R / DU):  $R[i+1] - P[i]$ [15] Release-to-Release (R2R / UU):  $R[i+1] - R[i]$ [15] Timing vectors are en-

riched with touch pressure, contact area, X-Y coordinates of the taps and spatial sensor telemetry (accelerometer, gyroscope) on touchscreen and multimodal signals on smartphones and tablets[1], [8]. Standard Evaluation Metrics Typical classification measures[4] are used to assess the performance of keystroke biometric models: When using a biometric system, False Acceptance Rate (FAR) is the percentage of unauthorized imposter attempts that are falsely accepted as valid attempts[5], [20]. False Rejection Rate (FRR): False impersonation attempts divided by the number of legitimate attempts[5], [20]. Equal Error Rate (EER): The point of operation where  $FAR = FRR$  and is one of the most important metrics of a system’s accuracy[5]. Authentication Confidence Score (ACS): A real-time trust score, normalized between 0.0 and 1.0, generated by risk engines that can be used to create dynamic access policies in Zero-Trust Architecture (ZTA) frameworks[18]. Benchmark Datasets The most commonly used public benchmark datasets are: 51 participants typing the password .tie5Roanl 50 times each, in 8 sessions (20,400 total samples recording 31 timing features)[12], [17]. In the Buffalo Keystroke Dataset, 148 subjects completed a combination of transcription and free-text typing tasks in 4 different keyboard models across 3 sessions separated by 28 days[4], [13]. Clarkson II Dataset: 103 subjects passively monitored over a 2.5 year period in a fully uncontrolled natural desktop typing setting[7]. Aalto University Dataset: A free text dataset of 136 million keystrokes from 168,000 desktop typists and 260,000 mobile typists [2]. We used 100 users from 20 countries, which also includes inter-key latency and handedness and demographic metadata[2]. SU-AIS BB-MAS Dataset: Multi-device behavioral dataset that includes keystroke data from 117 users using both desktop and mobile devices[14].

## 2.2 Review of Existing Research

The traditional machine learning methods and gradient boosted decision tree. The feature selection, outlier filtering, and ensemble decision-tree architectures are the key aspects of traditional machine learning research in keystroke dynamics that have been addressed extensively[16], [17]. Overall, gradient-boosted decision trees have demonstrated higher accuracy than some commonly used classifiers such as standard Support Vector Machines (SVM) and single decision trees on fixed text benchmarks[17]. On the CMU benchmark dataset, Muralidharan et al. evaluated multiple algorithms and showed that LightGBM outperform the rest of the algorithms with overall classification accuracy 94.68% (training time 14.47 seconds), better than the XGboost 92.89%, Random Forest 93.36%, SVM 84.85%, KNN 84.51%, and single Decision Trees 71.52%[17]. Likewise, Martins et al. used the fixed-text subset of the KeyRecs dataset, which they first filtered of anomalies using exploratory data analysis (EDA)[16]. They also found that LightGBM binary classification also has the best discrimination between genuine users and impostors with 80% F1-score and EER of 0.20%, FAR of 0.04%, and FRR of 0.36%[16]. The multiclass user identification over free-text data was addressed by Sharma et al. [13] who converted typing timing intervals into a multi-channel Keystroke Dynamics Image (KDI). A Random Forest classifier trained on flattened KDI representations has achieved the State of the Art identification accuracy of 93% with 148 users in the Buffalo dataset, demonstrating the ability of ensemble decision trees to outperform deep vision models that are trained on tabular behavioral features[13]. Deep Learning amp; Tempo-

ral Sequence Architectures Deep neural network architectures have been popularly used to deal with the long-term temporal dependencies and contextual variations in free-text typing: Lu et al. transformed the free-text timing streams into image-like transition matrices and trained CNN with Cutout regularization to avoid overfitting, and achieved 0.0078 EER and 98.53% accuracy on the Buffalo dataset[6]. To tackle this, Yang et al. created the TKCA framework which utilizes key-name vector embeddings and the hold and digraph flight times being passed to a Bidirectional LSTM (Bi-LSTM) network[7]. When tested on the uncontrolled Clarkson II database, TKCA obtained an 8.28% EER with short sequences of 30 keystrokes, and 2.78% EER with 190 keystrokes[7]. Al-Waisy et al. used a Bidirectional Recurrent Neural Network (BRNN) and Dynamic Weighted Dipper Throated Optimization (DWDTO) on mobile touchscreens for parameter tuning and obtained classification accuracy of more than 98.89%[8]. Acien et al. have proposed a Siamese Recurrent Neural Network (SRNN) architecture that has been trained on 168,000 subjects from the Aalto University dataset, demonstrating that deep temporal keystroke embeddings can be scaled to larger user bases of more than 100,000 users and can be trained with as few as 50 keystrokes per user[2]. Hybrid Deep-Ensemble Frameworks (LEKAM): Sasikumar and Nagarajan proposed a framework called LEKAM, which involves polynomial feature expansion, SelectKBest ANOVA F-score dimensionality reduction, LSTM temporal feature extraction and soft-voting ensemble (Random Forest, XGBoost, MLP) [19]. LEKAM reached a classification accuracy of 94.75%, EER of 0.0100 and inference throughput of 16092.93 samples per second on the KDA dataset and with the inclusion of SHAP (SHapley Additive exPlanations) for model interpretability[19]. Alazab et al. designed a multi-layered voting Deep Neural Network with a set of 170,000 users, with an accuracy of up to 99.7% and an EER of between 0.15% and 1.11%[4]. Zero-Trust Architecture (ZTA) Continuous Risk Protocols To incorporate continuous authentication in a Zero-Trust Architectures (NIST SP 800-207), real-time identity trust estimation without affecting the system responsiveness is required[15], [18]: The other multimodal Deep Fusion approach by Palavali et al. created a hybrid CNN-LSTM model combining the keystroke dynamics, mouse movement trajectories and the context of the workstation device to generate a real-time Authentication Confidence Score (ACS)[18]. It was tested with 60,000 enterprise sessions, and got results of 96.8% accuracy, 1.7% FAR, 1.5% FRR, an AUC of 0.985, and an inference latency of 240ms[18]. Reinforcement Learning Agents: Bansal and Ouda considered continuous authentication as a Markov Decision Process (MDP) problem and implemented a Double Deep Q-Network (DDQN) agent with Principal Component Analysis (PCA) feature encoding on the SU-AIS BB-MAS dataset[14]. The reward-driven DDQN agent attained an accuracy of 94.77% for training and 81.06% for testing (EER 0.0323)[14]. Lee et al. introduced Unconsciously Continuous Authentication Protocol (UCAP) that uses a hardware Trusted Platform Module (TPM) to protect session keys and continuous verification scores[15]. UCAP proved to be resistant to credential spoofing, replay attacks, and session hijacking through formal verification with Scyther and AVISPA[15]. Chauhan et al. created Continual Learning (ContAuth) to counteract the degradation in performance that is caused by natural variation in typing habits over time. To avoid catastrophic forgetting when new users enroll, ContAuth uses an LSTM feature extractor, combined with Stochastic Gradient Descent (SGD) online learning, and an iCaRL Class Incremental Learning (CIL) algorithm. Gener-

alization across Scenarios: Domain Adaptation Partial Password Verification One problem with the actual implementation is the variation in devices and keyboard masking requirements[11]: Cross-Keyboard Adaptation (CrossBehaAuth): Chen et al. proposed a CNN+GRU network combined with the temporal-aware learning and local Gaussian data augmentation method[11]. This system is capable of transferring user typing knowledge from a heterogeneous set of physical keyboards and from different recording sessions using only a small amount of calibration data[11]. Partial Password Verification: Giot et al. proposed a Siamese Neural Network (SNN) multi-shot classifier for verifying identity at partial password login (when online banking systems ask for a random subset of characters) to thwart shoulder surfing attacks[12]. The SNN framework was found to be greatly outperforming conventional statistical measures such as the Scaled Manhattan Distance[12]. Adhering to privacy preserving protocols and encrypted-domain computing. Following privacy-preserving protocols and encrypted-domain computing. As privacy concerns under GDPR and CCPA regulations are raised by continuous biometric collection, recent research focussed on cryptographic and decentralized learning techniques[10]: One of the other popular methods for privacy-preserving continuous authentication is Homomorphic Encryption Oblivious Transfer, which was introduced by Baig et al. [10]. This architecture allows for computing cosine similarity scores on encrypted biometric feature vectors without knowing what specific tasks had been performed by the user, or decrypting sensitive templates. Federated Learning Differential Privacy: Palavali et al. combined federated model updates with differential privacy noise injection ( $\epsilon = 2.5$ ,  $\delta = 10^{-5}$ ) to achieve 30% reduction in collecting centralized biometric data without compromising the accuracy of verification (95.8%) on enterprise endpoints[18].

## 2.3 Summary of Key Findings

The examination of these twenty studies reveals that continuous authentication has evolved from a point-in-time to an on-going, real-time, process of identity confirmation that is based on how a person interacts naturally with a touchscreen or keyboard [5], [20].

The literature is divided into two coherent themes at the feature-engineering level. In the case of fixed-text and short-password inputs, classical gradient-boosted ensembles (e.g., LightGBM and XGBoost) significantly outperform simpler classifiers (e.g., SVM, KNN, single decision trees) and are even superior to deep neural networks in terms of performance, with much lower training time (see [16], [17]). However, for free, unconstrained typing, deep temporal architectures – image-transition-matrix CNNs [6], Bidirectional LSTMs [7], Siamese recurrent networks [2], and hybrid CNN-LSTM ensembles [18], [19] – are required to capture the long-range contextual dependencies in the typing rhythm of natural typing, and these architectures scale significantly better to large, uncontrolled datasets from a few hundred subjects [7] to the 168,000-user Aalto corpus [2].

Another recurring theme is that there is no one single behavioural signal. Combining keystroke dynamics with mouse trajectories [9] or touchscreen pressure and gesture information with the password [1], [8] or with device and session context [18] consistently reduces error rates compared to keystroke-only baselines while increasing data-collection overhead and cross-device complexity, the problem tackled

by calibration-transfer techniques across heterogeneous keyboards [11] and partial-password Siamese verification [12] respectively.

Third, the field is increasingly moving towards embedding continuous authentication within Zero-Trust Architecture (ZTA), moving away from a one-off decision on whether to accept or reject a request, and towards a continuously updated trust decision: an Authentication Confidence Score [18] or a hardware-backed trust protocol using a Trusted Platform Module [15]. This is further extended in framing the problem of authentication as a sequential decision process solved using reinforcement learning [14] where the system dynamically adjusts its level of confidence instead of a fixed classification threshold.

Fourth, privacy and long-term robustness are still unsolved problems. Since the continuous authentication relies on continuous collection of the behavioral data, several studies introduce privacy preserving computation techniques: homomorphic encryption and oblivious transfer [10] and federated learning with differential privacy [18] are proposed to prevent the centralization of sensitive biometric templates. Type behavior is not constant, however; as the user gets tired, hurts, or practices over time, typing patterns change, which will reduce the accuracy of the typed models, and it will be impractical to train the model from scratch for each new user that joins it [3].

Finally, no single benchmark or evaluation protocol is standardized across the reviewed studies: they can be as small as less than 100 users or as large as more than 168,000 users, have fixed or free text, require desktop or laptop or mobile touch-screen devices, and report accuracy, EER, FAR, or FRR — making cross-study comparisons difficult, as indicated by several surveys in this set [5], [20]. Together, these results suggest the need for a consistent and all-encompassing multimodal, adaptively-learning framework with a unified performance metric, which is the focus of this thesis' DDQN-based multimodal fusion approach.

# Chapter 3

## Conclusion

In conclusion, we can say that the concept of point-of-entry authentication like passwords and PINs, is not safe for modern life styles because it struggles to identify the actual identity of the user and can be the equal for the whole session. As Continuous Authentication passively monitors behavioral biometrics it effectively closes the trust gap. Like, the motion of users typing, the movement of the mouse and how the users swipe the screen. It verifies the user's identity on the fly. Those journals depict that some sort of AI models, such as deep learning model, Reinforcement Learning model(which is reward giving) can accurately identify the exact user. The accuracy often crosses over 95%, when it learns these unique behavioral fingerprints. Also, here in research multimodal is used to combine multiple behavioral fusions, which helps to lower the risk of an attacker from hijacking information or any misdeed, because it is quite impossible for an attacker to copy those multiple behaviors together. Though the security makes security easier for the users, users also have to think about "Model Drift", that happens when a user's health gets worse and behaviour changes because of aging, stress, tension etc. So, to preserve those important and sensitive information of users, there we can see the use of privacy tools, such as homomorphic encryption and on-device processing, which make sure that the personal information of the users will be safe and never be stolen at any condition. This study demonstrates that behavior-based continuous verification is an essential step toward a safe and user friendly "Zero Trust" future by lowering the "exposure time", the duration of time the device gets exposed after being unlocked and so that it can be close to zero.

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